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Data Report: Flooding Remediation

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Project Summary

The primary goal of our project is to identify areas along Mass Ave that are vulnerable to flooding or have previously experienced flooding. Through thorough data analysis and visualization, we will pinpoint these high-risk locations and propose suitable tree-planting sites as a means to mitigate water runoff. Our project's scope extends beyond flood mitigation. By analyzing the data, we also aim to identify other community areas that could benefit from tree-planting initiatives. This approach allows us to address not only flood-prone locations but also find opportunities to enhance the overall environmental quality and resilience of the Mass Ave Corridor.

Our project leverages an array of data sources, drawing upon datasets provided by our professor and teaching assistants, as well as externally sourced data. Notably, we incorporated the Speak for the Trees dataset and the 311 Service Requests Dataset_[1], which were foundational for our analysis. To gain an understanding of the weather conditions on days featuring cases from the 311 dataset, we utilized precipitation data from the National Oceanic and Atmospheric Administration spanning 2011-2023_[2].

In addition to this empirical data, we enriched our understanding of our project's context and implications by reviewing academic research on the role of trees in flood mitigation_{[3][4][5][6]}. This literature provided key insights into how trees can limit flood damage and offered specific information regarding the volume of water an average tree can absorb, dependent on its diameter. Our analysis benefited from these findings, informing our data interpretation and guiding our investigation into the synergistic relationship between urban forestry and flood management.

As aforementioned, there have been studies pointing to the conclusion that urban flooding and soil erosion can be mitigated by trees. Given the rise in storms and unpredictable weather patterns associated with climate change, flood mitigation is increasingly crucial for urban cities, particularly those located near coastal areas. This leads us to our main motivating question that we hope would assist our community partners, Speak for the Trees and Mass Ave Coalition with their endeavors - Can trees reduce damage caused by floods? In addition to this question, we also hoped to answer in our project the following: Where historically have there been floods? Where in Mass Ave is there a higher risk of

flooding? Where do you think the best places to plant trees would be? Provide information to the user about the typical age, size, and water absorption rate of a specific tree, and how much water the trees on Mass Ave absorb.

Data

To achieve our goals, our project focuses on analyzing relevant data that provides insights into the types of 311 cases reported, with a particular emphasis on the Boston Water and Sewer Commission and the Mayor's 24-Hour Hotline. The project also looks into rainfall and elevation data as supplementary data sources that will help drive discovery.

311 Data

We will be using the Boston Water and Sewer Commission and Mayor's 24-Hour Hotline which provides us with the 311 data. This includes all service requests made by the Boston community from 2011-2023. This allows us the opportunity to concentrate on specific parts of the data such as the Subject column of the service requests, which includes information on catch basin issues, manholes, potholes, and more, to gather data that indicates areas prone to flooding. This data contains strings describing issues, dates, date-times describing the creation and other relevant data, and numerics describing latitude and longitude and other values.

Furthermore, through filtering the data by neighborhood/location, we can identify specific areas along Mass Ave where flooding incidents have occurred or are likely to occur in the future. This information will be instrumental in determining suitable locations for tree planting as a means to mitigate water runoff and prevent floods.

This data was very large and clunky to deal with as there were a lot of vestigial components that were unneeded, so most of the time was spent carefully removing what had no value.

We mapped out all of the columns assigned in the 311 Data and eliminated any columns we believed were not relevant to our end goal. We then detailed and marked what specific groups within each column we should focus on, for instance, "Pothole Repair" or "Sidewalk Repair" under the "Type" column. After removing unnecessary columns, we realized that a few of the rows under "Location_zipcode" were null values. However, we had latitude and

longitude data which we were able to use to fill in the zip codes. We also decided to make a new column, "Year", to simplify our data visualization process

Elevation Data

We had initially decided to also work with elevation data collected from the Massachusetts government website MassGIS, the state's database for interactive maps and related descriptive information. From their MassMapper tool, from the right-hand tool, we followed the drop-down from physical resources to Topography contours. Under this, we selected the *3m contour labels feet* ([Figure 7](#)).

After parsing through the data and conducting hours of EDA, we found it very difficult to convert the topographic lines representing elevation to static data points for each coordinate. So we had to find another dataset. After exploring a bit, we came across the Google API for elevation data. After creating a new project and generating an API key, we realized that this dataset focused on much larger latitudes and longitudes. Only two such coordinates passed through Boston, we weren't able to get enough data to graph or work with for correlation. Attempts were made under the file 'Google API Elevation.ipynb' found under the folder "Ananya - Elevation" in our project folder. This is why we ultimately decided to not work with elevation data.

Rainfall Data

We will be using data from the National Oceanic and Atmospheric Administration, which provides precipitation data for various areas in Boston. This dataset dates back to the 1800s but we are using the data from 2011 - 2023 for relevance. Access to historical rainfall data in our specified region of interest will help identify patterns of heavy rainfall events and understand the potential risk of flooding ([Figure 5](#)). This will help us with the task of examining the history of flooding, looking at flooding data for Mass Ave.

The data clean-up process for this dataset was relatively simple. The dataset had a few null values that we decided to remove because averaging out these variables can lead to misleading results by pulling the imputed values toward extreme values. Other than the null precipitation values, everything else in the dataset was fine and did not need any adjustments. The data contains a String representing a Station, a String representing the

name, an int for Latitude, an int for Longitude, an int for Elevation, a Date in date format, and an int for PRCP (precipitation).

Speak for the Trees Data

For our project, we will utilize the "Speak for the Trees" dataset, kindly provided by the teaching assistants, as a foundation for creating interactive, information-rich tree graphs. During the data cleansing phase, we initiated a two-step filtering process. Firstly, we eliminated all incomplete (NaN) records and discarded unnecessary columns from the dataset.

Secondly, we narrowed down the tree locations to align with the precise coordinates of the Mass Ave section, which is the focal area for our report. To achieve this, we used Google Maps to identify the desired section of Mass Ave, then defined a box-like boundary around it. We divided this box into a 3x3 grid to enhance precision, enabling us to gather specific data for each smaller sector within the overall Mass Ave region. We found it helpful to divide the selected Mass Ave Corridor into sectors since we were working with 2D Coordinates, it was difficult to work with area data ([Figure 6](#)).

This granular approach will thoroughly compare various cases against the tree populations within each subsector. The insights derived from these comparisons will be instrumental in designing an interactive graph, enriching our final submission.

Sewer Data (Bonus Dataset)

We decided to go out and gather data about the number of sewers along the Massachusetts Ave Corridor. The data we collected was crucial for our project as it helped us in identifying areas vulnerable to flooding or even correlate troubled areas where there have been flooding in the past to the number of sewers in the area. Sewers play a substantial role in managing water runoff so understanding the placements and distributions of these sewers was important to locate areas that were more prone to flooding and where tree planting could be beneficial (which contributes to answering our main motivating question).

The data we collected had four columns: a boolean to denote whether or not there was a sewer in the area, coordinates of the sewer, the number of sewers in the area, and the location of the sewer (whether it was located on the road or sidewalk). The coordinates allow for spatial analysis and visualization, while the number of sewers provides insights into the density of infrastructure in these areas.

To gather the data, our group went out and walked down the Mass Ave Corridor physically observing and documenting the presence of sewers, recording the coordinates using the Google Maps feature, and noting the number of sewers and their respective locations. The most challenging part of the data-gathering process includes ensuring consistent and accurate documentation of sewer information since we needed at least one member of the group to walk down the other side of the sidewalk and communicate sewer information from across the street.

From the data gathered, our group gained insights into the distribution and density of sewers along the Mass Ave Corridor. We have identified areas with higher concentrations of sewers and those lacking adequate sewer infrastructure ([Figure 8](#)). This information can help in assessing the flooding risks of different blocks along the corridor. Additionally, this information allows us to highlight specific areas where tree planting could be beneficial due to the lack of sewer infrastructure. The data-gathering experience provided our group with a more comprehensive understanding of the project's scope and challenges. It allowed us to develop insights based on observed conditions. Overall, this firsthand experience enhanced our group's perspective on the project.

Observations and Insights

Key findings from our reports have revealed a few significant trends. Over the past few years, we have observed a rising trend in unresolved cases, with an increasing number of issues left open. Furthermore, there's an escalation in the total count of cases reported annually. Particularly noteworthy is the trend that certain neighborhoods within Boston consistently report a higher frequency of problems compared to others. In our pursuit to decode these patterns, we plan to integrate data from the 'Speak for the Trees' organization. We aim to examine whether there exists a correlation between the type and density of tree populations in these areas and the type of issues frequently registered in

the 311 datasets. This intersection of data may reveal fascinating insights into the nature of the issues and contribute to proactive urban planning solutions.

From our visualizations, there were a couple of interesting things that were observed. The visualizations will be below, for reference. In [Figure 1](#), which is the density distribution of open cases across Boston, one can see that the Back Bay/Roxbury region has an unusually high density of open cases in the area. Since this is our target area, this is important to reference and keep in mind. There are only two regions that are comparable to the density of open cases, those being the North End and Faneuil Hall regions. Since all three of these are historically older regions, it is possible that a lack of adequate infrastructure is contributing to the number of open cases. [Figure 2](#) demonstrates that the sum of open cases is increasing consistently over the years, which suggests that the issues are not being acted upon in a timely and appropriate manner. The cause of this could be many things, from crumbling infrastructure making upkeep more difficult to lack government maintenance on these topics, causing the backlog to balloon.

[Figure 3](#) dives a little more in-depth into this, and touches upon the cases completed by year and open cases by year. There have been fewer and fewer cases completed after 2020 despite a growing number of open cases. This could suggest that the current resources allocated to solve these problems just aren't enough. [Figure 4](#) is a tree map of the count of open cases by type. It seems that pothole repairs, an internal issue, are taking up the bulk of the cases and causing issues. It could be that these issues are disproportionately labor-intensive and take up resources. [Figure 5](#) is a plot created using the precipitation data denoting the amount of rainfall per year in the Boston area from 2011 - 2023. [Figure 8](#) is a visualization we created using the Sewer Dataset that we gathered. We can use this information to later determine areas that lack draining infrastructure and highly absorbent trees to suggest the most optimal areas that could benefit from planting more trees.

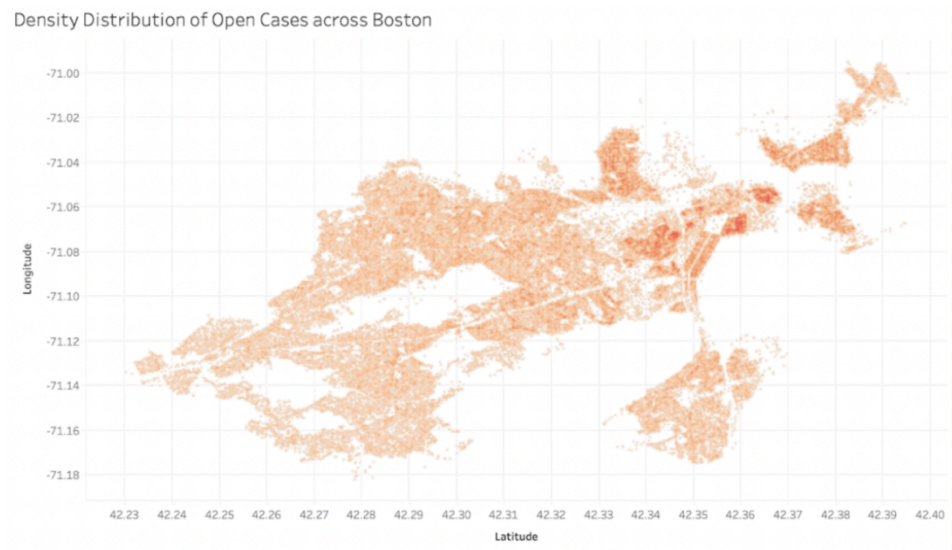


Figure 1: Density Distribution of Open Cases across Boston

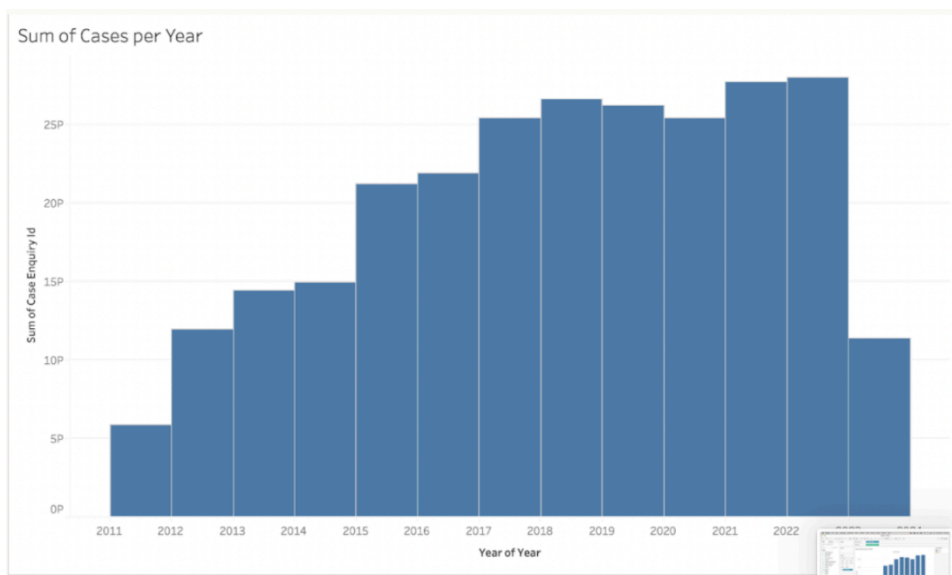


Figure 2: Sum of Cases Per Year

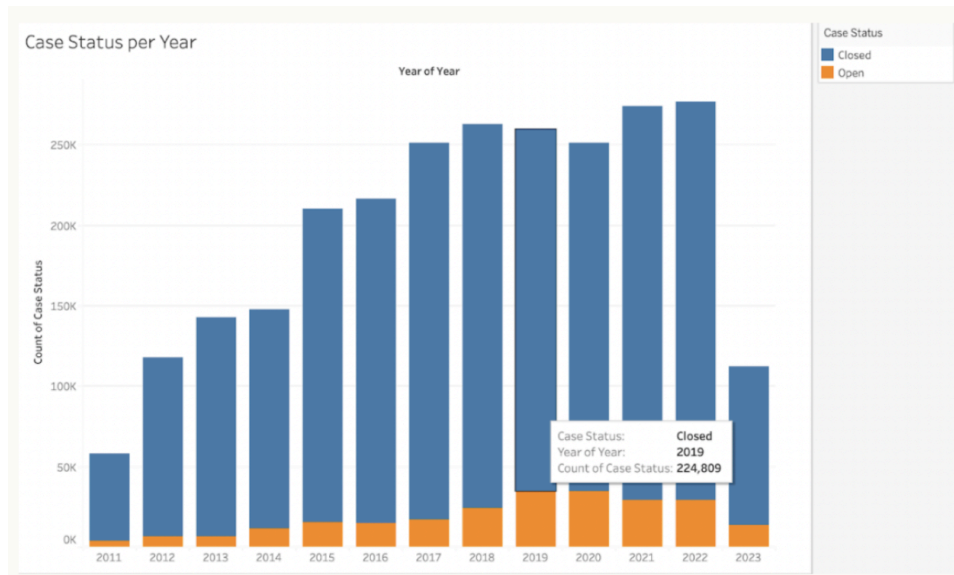


Figure 3: Case Status Per Year

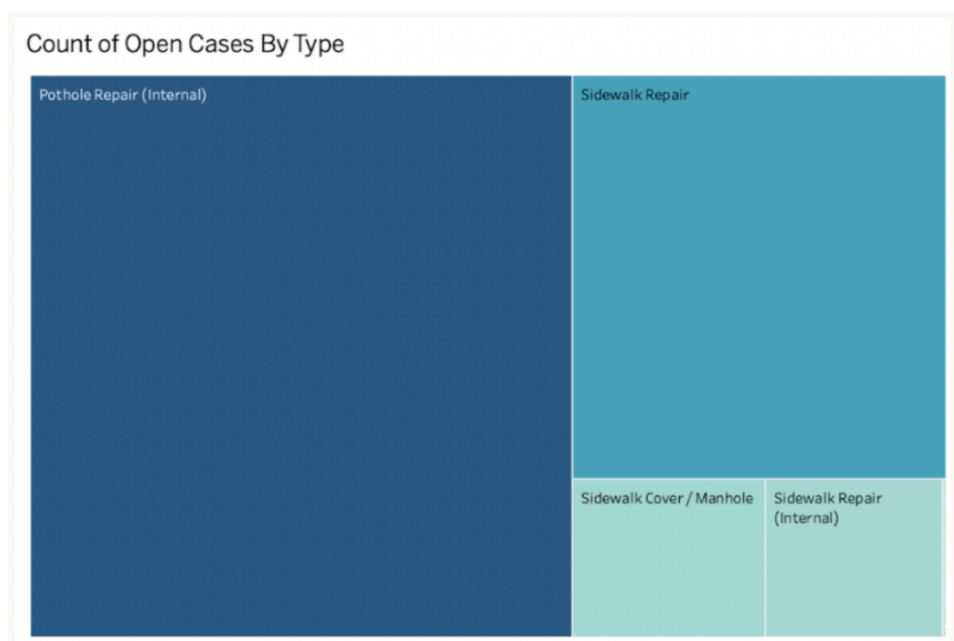


Figure 4: Count of Open Cases by Type

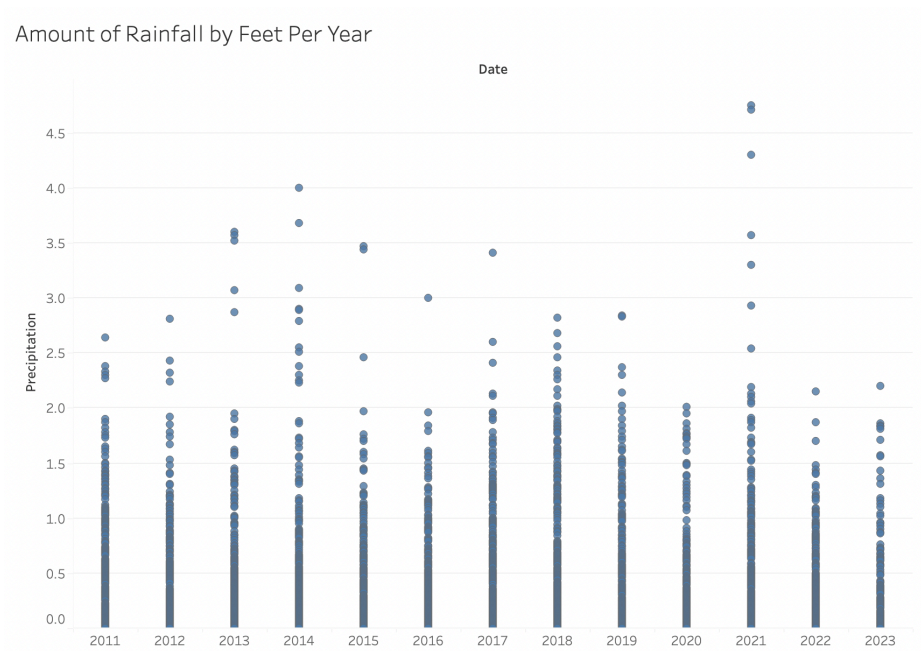


Figure 5: Amount of Rainfall by Feet Per Year

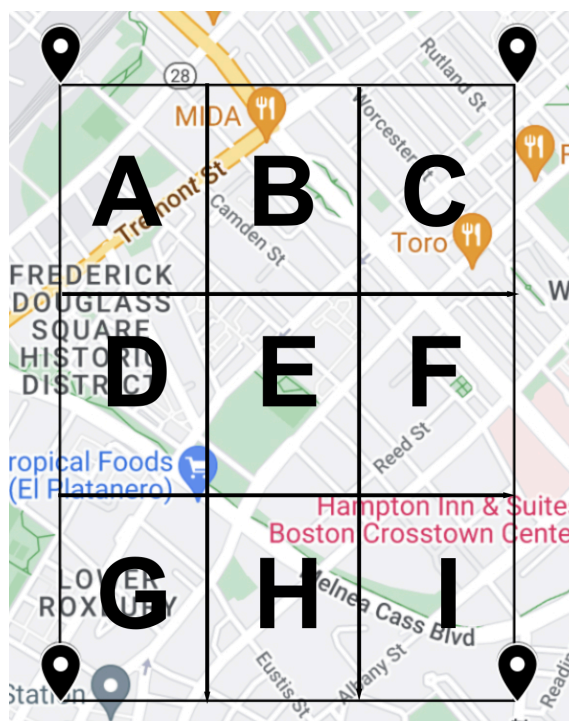


Figure 6: Sector Map

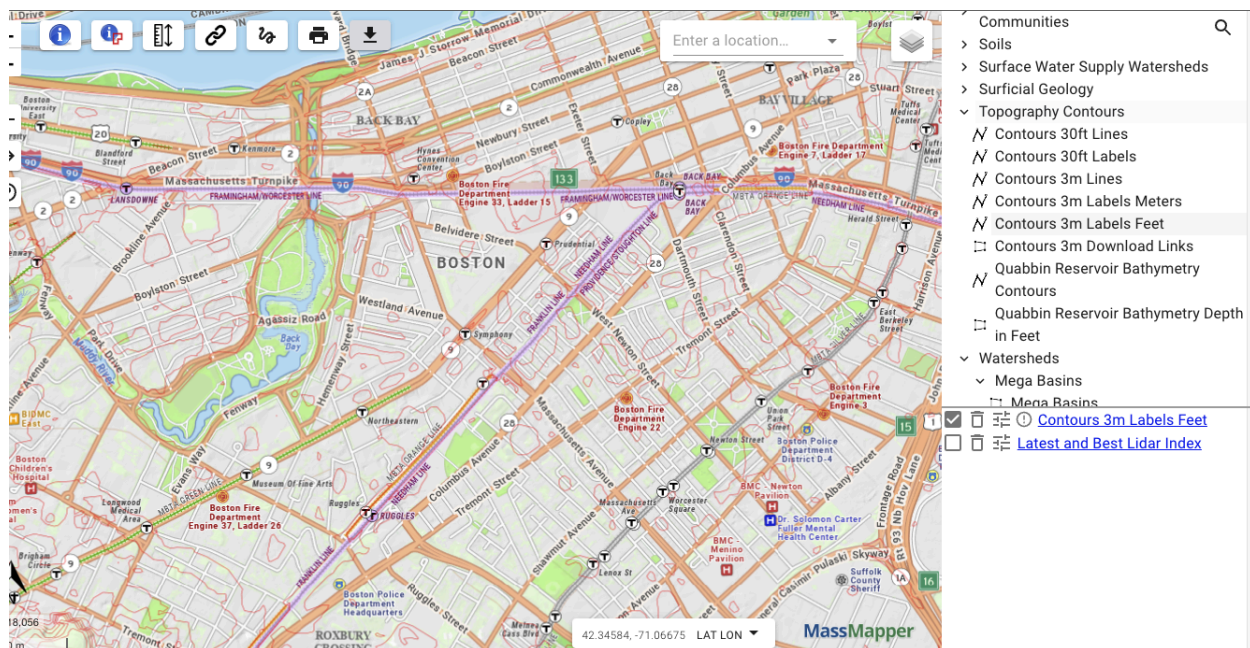


Figure 7: Elevation Data

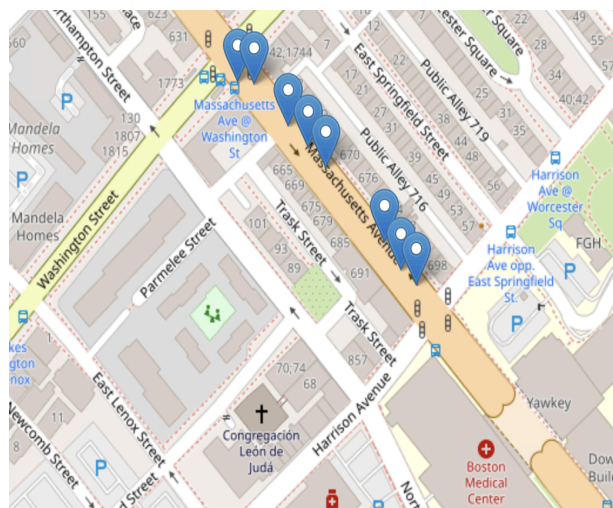


Figure 8: Sewer Data (Bonus Dataset)

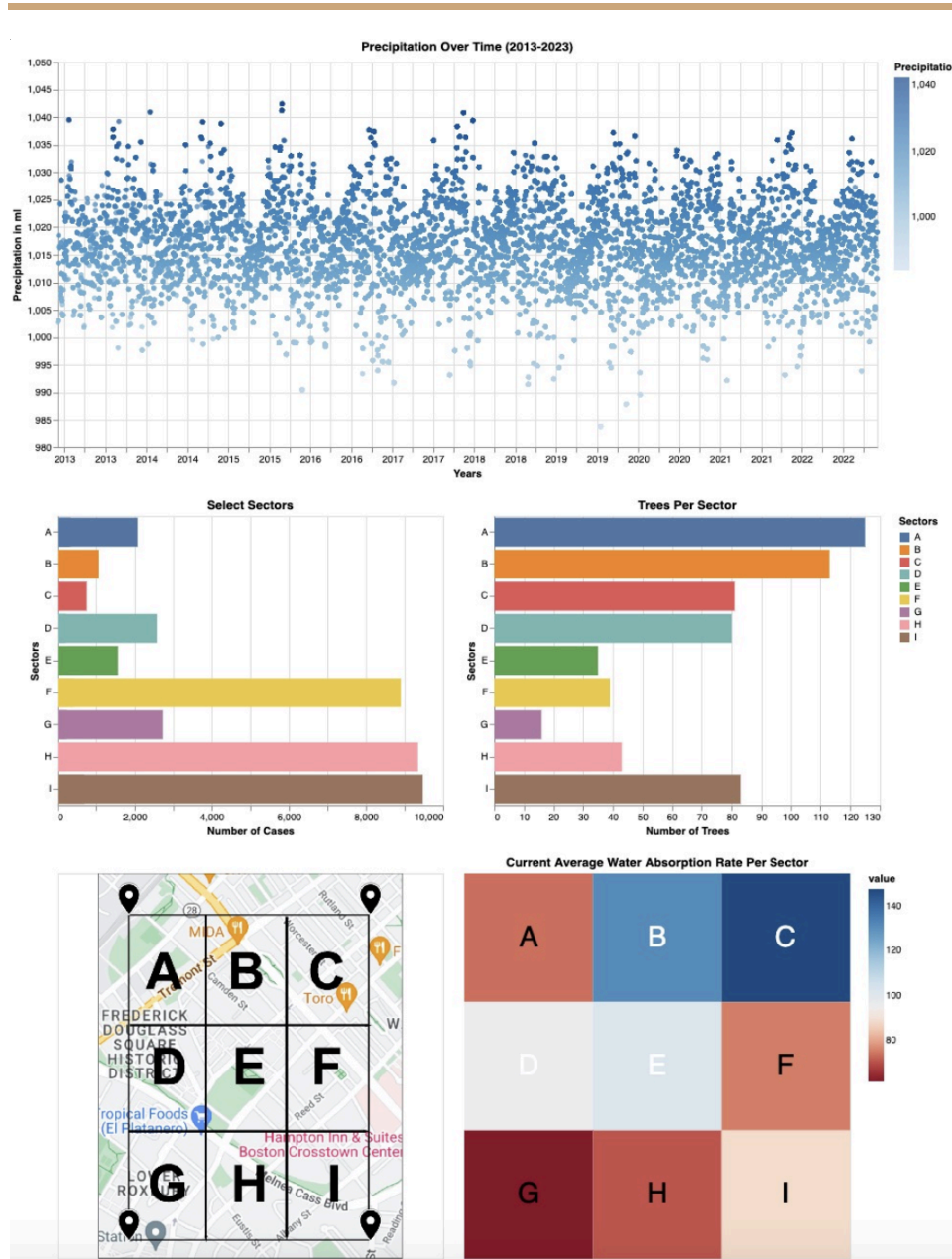


Figure 9: Interactive Figure (Static Grab)

The interactive figure helped us understand a few more details regarding the scope of our project. This figure mapped out the precipitation over time in Boston and linked it to the cases reported per sector in Boston. Furthermore, it also represented the count of trees per sector when highlighted. This was a very complex diagram so we decided to divide Boston into target [sectors](#) based on coordinates.

To show more of a correlation between trees and helping reduce flood risk we decided to divide Mass Ave into different sectors and also choose a few areas out of the site. This will allow us to compare each sub-sector with another.

Our interactivity includes a brush that allows the user to select a specific area of the scatter plots containing cases according to precipitation data for that day. This will then show data for those cases in the next bar chart showing the number of cases per sector. We also have a drop-down selector that allows us to select a specific sector which shows the case data in the scatter plot and also the cases per sector in the bar chart. Additionally, we have created a water heatmap grid to show how effective each sector is at water absorption compared to the other sectors.

This information can aid Mass Ave Coalition to see how effective each sector is at reducing flood risk. We found out that the more trees are found in a sector, the fewer cases were reported. In turn, Mass Ave Coalition can use this information to effectively assess where tree planting can be beneficial along Mass Ave.

Task Analysis

This visualization will be primarily developed to both explore and present the data. We will first discover the best places to implement change and then present this to our stakeholders. The primary stakeholders of the visualization will be the Mass Ave Coalition, local residents, and city planning authorities. The goal is to provide clear, insightful, and actionable information to the stakeholders, enabling them to understand the role of trees in preventing water erosion on Mass Ave in Boston. The visualization will help them identify areas at risk of flooding, areas where floods have occurred, and areas where tree planting could be beneficial in reducing water run-off or flooding.

Index (ID #)	"Domain" Task	Analytic Task (Low-level, "Query")	Search Task (Mid-level)	Analyze Task (High-level)
1	What is the current layout of trees on Mass Ave?	Retrieve Values	Locate	Discover
2	What does the Mass Ave layout look like? Where are the drains, sprinklers, water retention bags, and other relevant structures?	Retrieve Values	Locate	Annotate
3	Examining the history of flooding, looking at flooding data for Mass Ave.	Compute Derived Values	Lookup	Discover
4	Identifying the areas in Mass Ave most prone to flooding.	Extremum	Explore	Discover
5	Examining and analyzing the data from the 311 hotline.	Retrieve Values	Explore	Annotate
6	Researching the type of trees that are most efficient at preventing run off.	Extremum	Explore	Derive
7	Examining the effect of trees on flood prevention. What attributes of a tree - type, age, width of trunk, impact its ability to prevent flooding?	Correlate	Explore	Derive
8	Understanding correlation of elevation of the road with flooding	Correlate	Lookup	Derive
9	Potential Tree planting arrangement to reduce risk and damage from floods	Determine Range	Browse	Present
10	Understanding the effect of flooding on the community.	Correlate	Explore	Discover

Speak for the Trees App Prototype

Upon scanning the QR code, users would be presented with a visualization that illustrates how much **water runoff that tree absorbs**. This is calculated based on the diameter of the tree. Using this information, we also calculate the amount of runoff the tree prevents in cubic meters.

The created half donut chart for the absorption amount by the trees helps users visualize the relative quantity of water. The aim of this visualization was to show users the 'impact' that tree has on any type of water runoff (in turn, flooding events).

Here are the visualizations we produced:

This is a Northern Red Oak
It absorbs: 205.00 Gallons

This tree prevents 0.60 cubic meters of stormwater runoff.



This is a Japanese zelkova
It absorbs: 54.00 Gallons

This tree prevents 0.60 cubic meters of stormwater runoff.



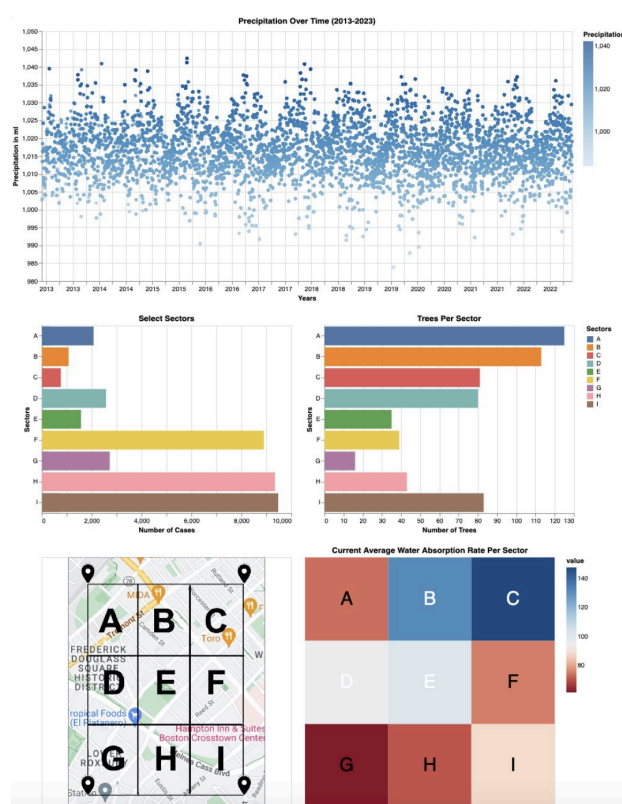
This is a Swamp White Oak
It absorbs: 237.00 Gallons

This tree prevents 0.84 cubic meters of stormwater runoff.



Conclusion

In summary, a thorough examination of the sector map reveals that Mass Ave is a complex blend of sectors A through F. Our detailed analysis of all the 311 case data over the years from these sectors establishes a crucial relationship: sectors with a higher tree count consistently report the fewest number of cases. This corroborates our thesis that an increased tree presence plays a significant role in mitigating damage caused by floods and water erosion.



Further proof of this is presented in the data we can see for Sector F. This sector records one of the highest case counts over the year, paralleled by a remarkably low tree count. Further analysis of the water absorption rates across sectors, as demonstrated by our heatmap, reveals that Sector F's rate is below average. This underscores the heightened vulnerability of this sector to flood and erosion damage, which explains the extremely high number of cases the sector has had over the years.

Turning our attention to [Figure 8](#), which captures the sewer data for Sector F, we observe a distinctive anomaly. This sector boasts a high number of road sewers but is

underserved by sidewalk sewers, potentially exacerbating the flooding issue. Given its deficient tree count and the supportive absorption data, we can confidently assert that an increase in tree density can significantly alleviate the flooding incidents in this area.

Further Work

In terms of further work, several enhancements are recommended to refine our analysis and deepen the understanding of the dynamics we're investigating. Firstly, the interactivity across all charts can be expanded to facilitate more cohesive and comprehensive data interpretation. The ability to draw connections and discern patterns across various visual representations of data would undoubtedly improve our insights.

Additionally, incorporating more time series data could provide a clearer picture of long-term trends and fluctuations, thereby enhancing the predictive power of our model. The application of tree species-specific datasets is also recommended, enabling us to account for the varying water absorption rates among different species and observe how they change over time according to their age and size. This dataset would allow us to make the second bar chart completely interactive with the cases data from 311, as we would be able to filter the trees that were present during the case.

Lastly, securing an elevation dataset for the roads in and around Boston would be invaluable. This would allow us to consider the role of topography in our analysis, thereby offering a more nuanced understanding of how water is distributed and absorbed in different areas. By factoring in these variables and enriching our dataset, we hope to create a more robust and reliable model for predicting and mitigating flood damage.

Works Cited

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